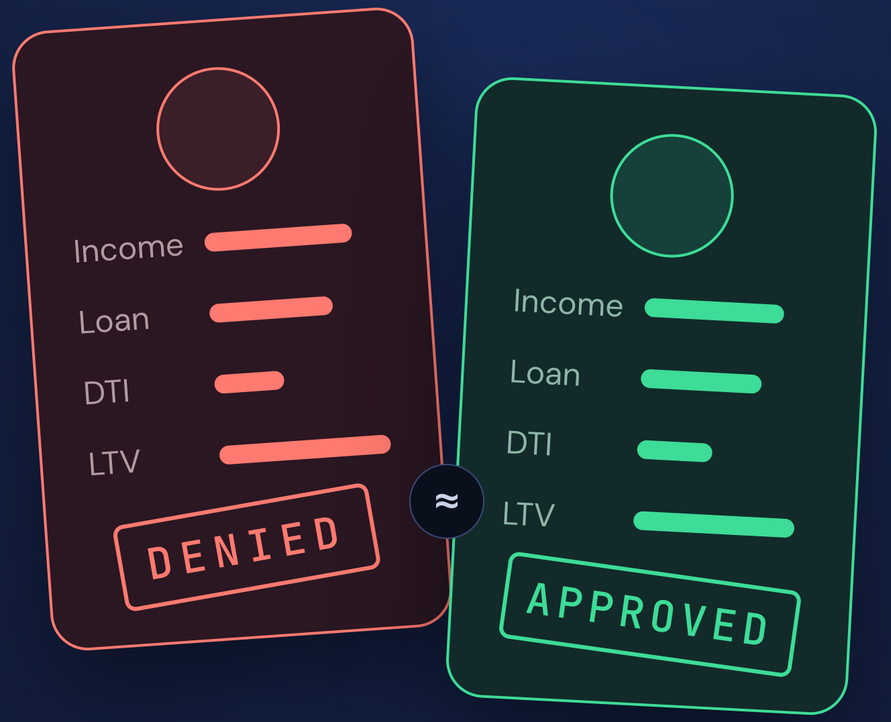
IBM Z DATATHON 2026 · REAL-TIME AI FOR CRITICAL DECISIONS

SecondLook

Same qualifications. Different decisions. Who notices?



TEAM WISE

AI FOR GOOD

ADVISORY ONLY

THE PROBLEM

Two applicants. Same profile. Different decisions.

Who checks why?

DENIED APPLICATION	
Income	$141K
Loan amount	$615K
Debt-to-income	46%
Loan-to-value	95%

DENIED

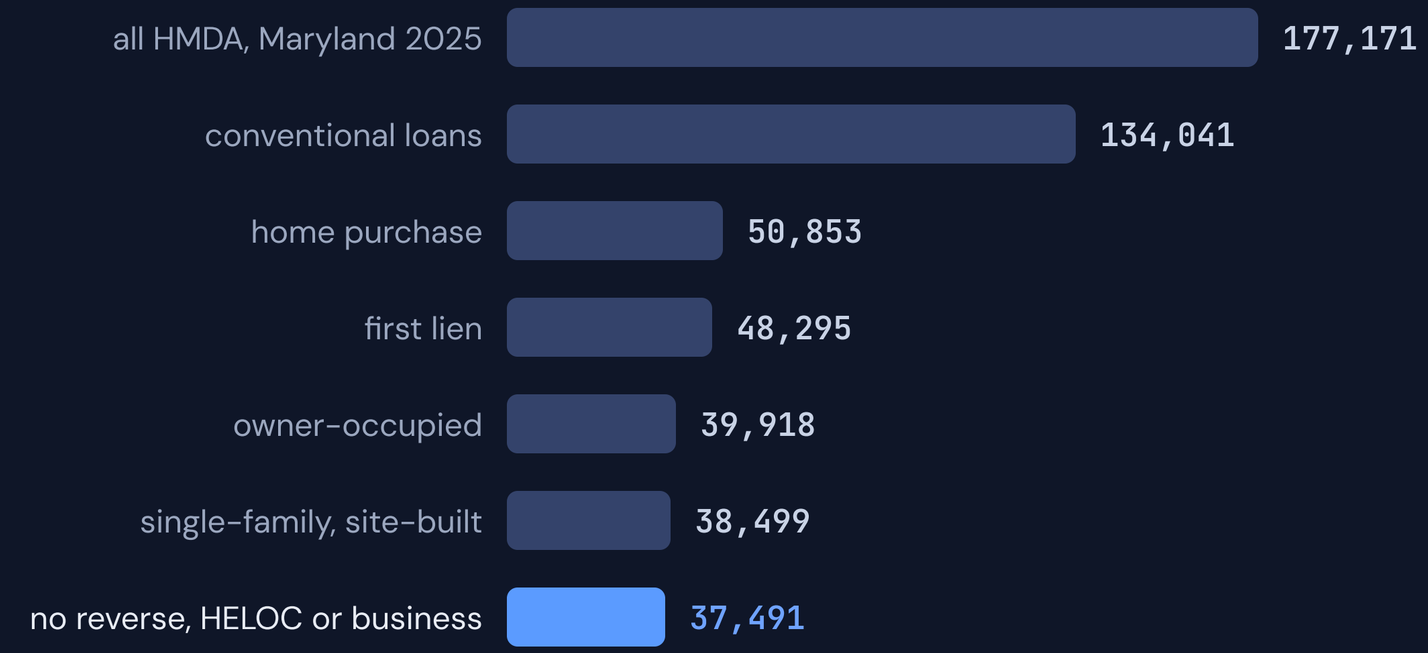
≈

APPROVED TWIN · SAME LENDER	
Income	$146K
Loan amount	$595K
Debt-to-income	45%
Loan-to-value	95%

APPROVED

THE DATA

From 177,171 records to 37,491 comparable applications



34,891
approved (originated or approved, not accepted)

2,600
denied

447
lenders

One product family, so every approved and denied application is judged on the same footing.

SOURCE: HMDA LAR 2025, FFIEC · MISSING: INCOME 0.8%, DTI 0.6%, LTV 1.0%

THE SOLUTION

Match first. Explain second. A human decides.

01
Financial Twins Matching Engine
Find comparable applications

→

02
Structured Case Evidence
Financial differences, denial reasons, missing information

→

03 · LINUXONE
IBM Granite
Explains, then every sentence is checked against the facts

→

04
Human Reviewer
Verifies the evidence and decides next steps

STAGE 1 · MUST MATCH EXACTLY

same lender | conventional | home purchase | first lien | owner-occupied | site-built home

STAGE 2 · HOW CLOSE? (TOLERANCES)

income ±15% | loan ±10% | property ±10% | DTI ±5 pts | LTV ±5 pts | term ±60 mo

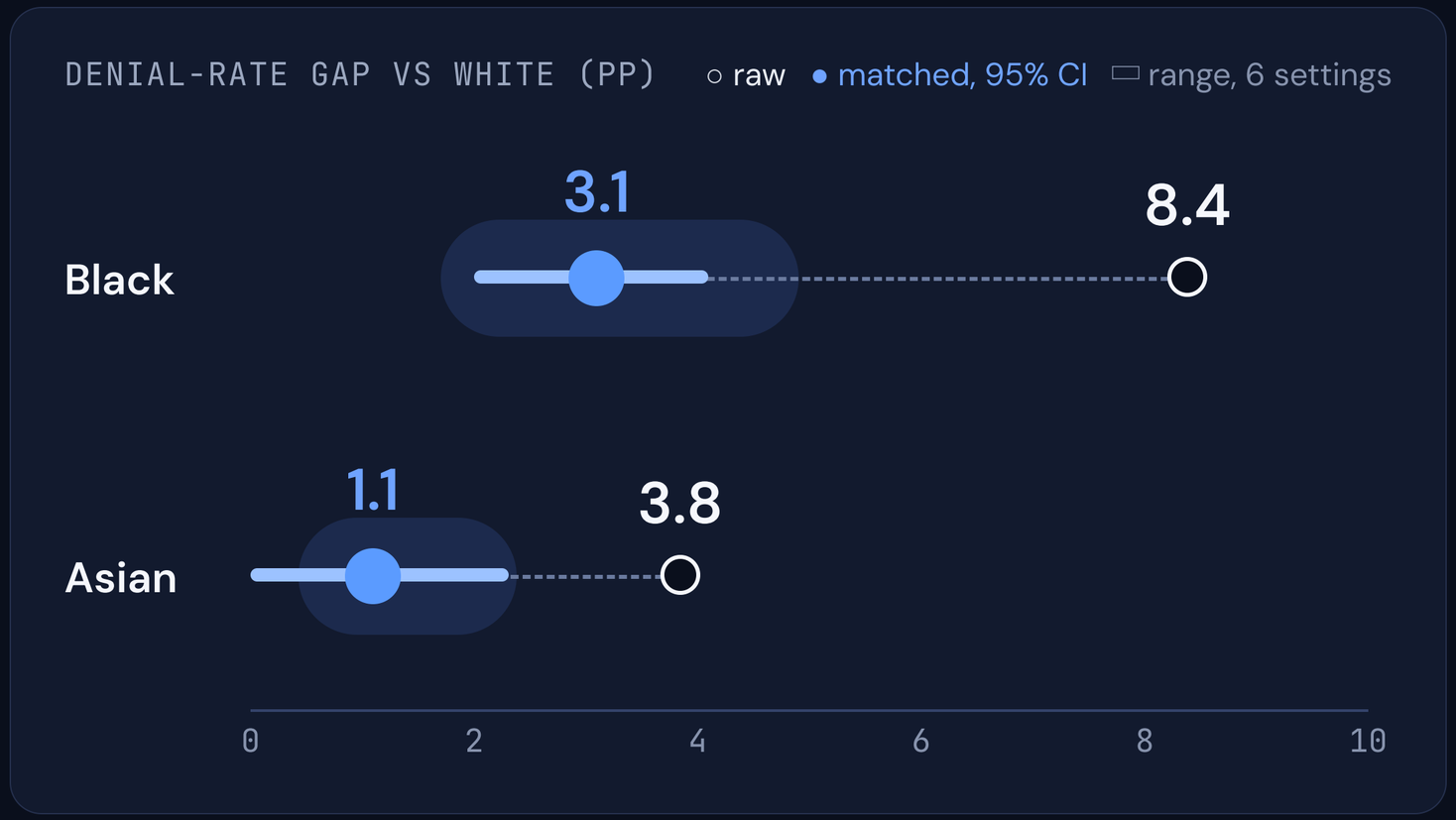
A similarity score, not a probability of approval.

Blind by construction
Race, sex, ethnicity and age live in a separate table, joined only after matching.

DENIAL REASONS ARE SHOWN TO REVIEWERS, NEVER USED FOR MATCHING

BLIND REVIEW → REVEAL

Match on finances first, then reveal



An observed disparity. Not proof of discrimination.

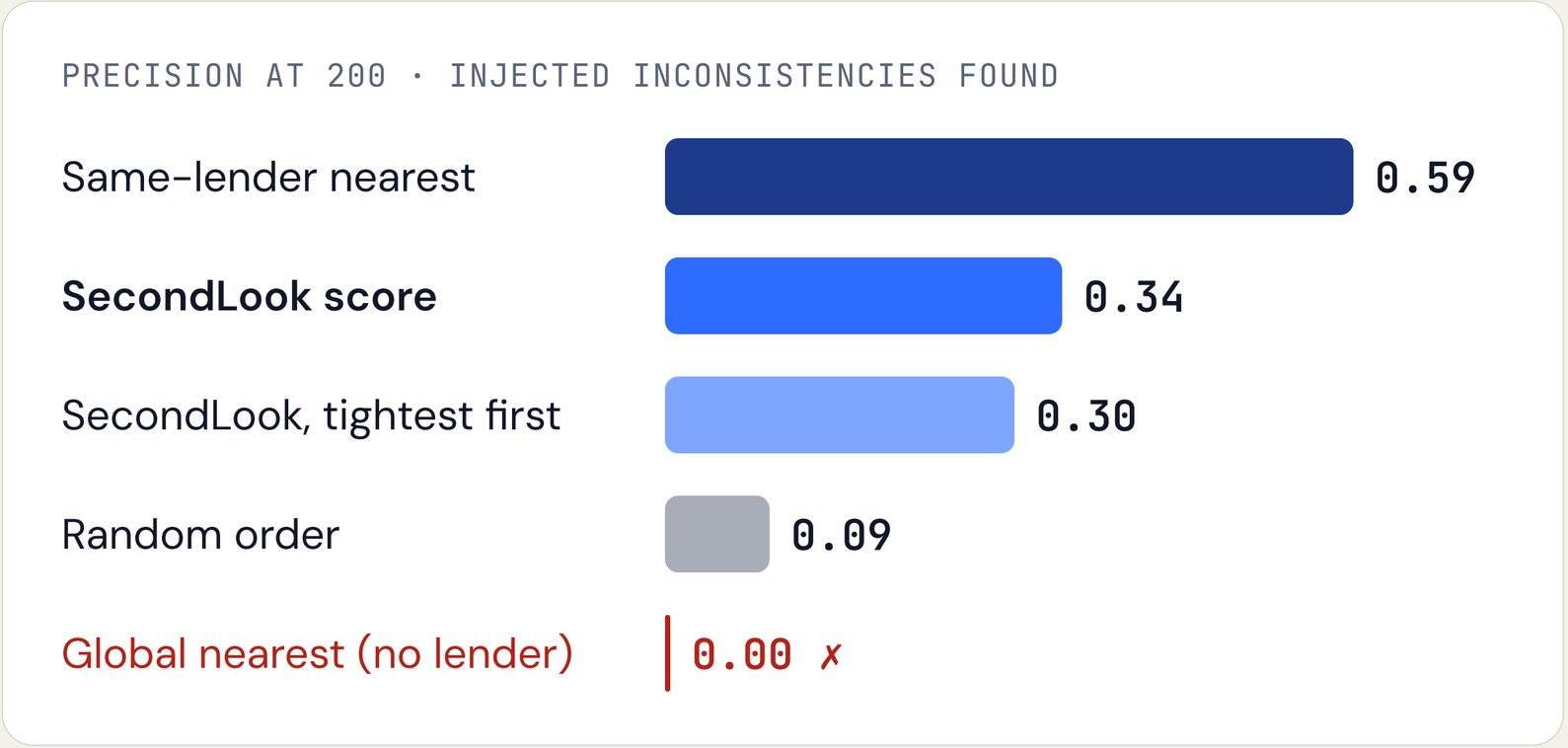
Strong finance matches exist for 69% of Black and 59% of Asian applicants.

Black gap ranges 1.7 to 4.9 pp across settings. Asian interval touches zero.

Credit scores are missing. Finances and county partly predict race (AUC 0.68 and 0.74), so “blind” is not proxy-free.

VALIDATION · SYNTHETIC TEST

An honest test: the lender constraint matters



Same lender is essential

Plain similarity ranks as well. No accuracy win claimed.

We add evidence and workflow

Tolerances, missing-data handling, reason check, audit trail.

Not measured: review time and real workload savings. Real wrongful denials have no ground truth in public data.

IBM LINUXONE · S390X · MEASURED, NOT CLAIMED

One controlled platform for a sensitive workflow

HMDA
SHED · ACS

LINUXONE · S390X · 3 vCPU · CONTAINER + SERVICES

Twin matching | Blind audit | XGBoost | Granite 4.0

Reviewer
SSH tunnel only

THROUGHPUT (REQ/S) VS CONCURRENT CLIENTS



~4 ms
median live twin search, simulated application

identical
result digests on s390x and on an arm64 laptop

3 workers
~118 MB each, systemd services, auto-restart

token-gated
demographic endpoints; no personal data in logs

Straight talk: not faster than a laptop (0.68 s vs 0.38 s for same-lender matching). zDNN initialises, but offload was no faster, so we claim no acceleration.

SECONDLOOK · TEAM WISE · IBM Z DATATHON 2026

SecondLook doesn’t decide who deserves a mortgage.

It makes sure decisions worth questioning don’t disappear into millions of records.

PUBLIC DATA

ADVISORY ONLY

RUNS ON LINUXONE

github.com/shreyvivek/WISE_Datathon

NOT IN THE 3-MINUTE VIDEO

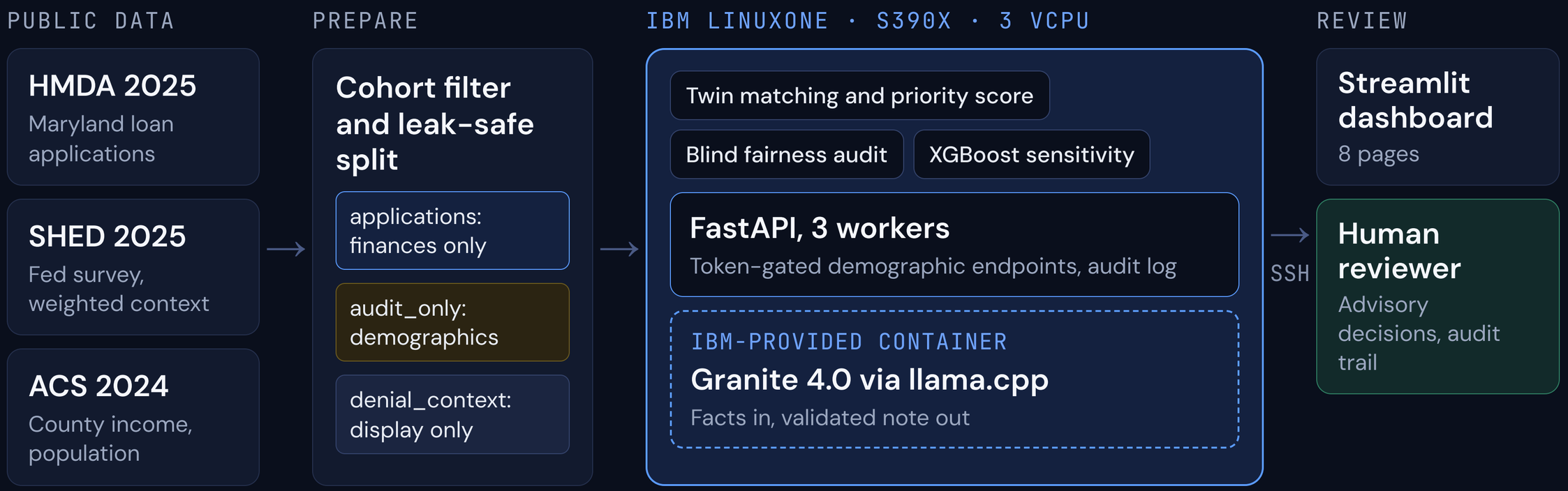
Appendix

Architecture, design flow and extra analysis for judges, the Q&A and the written submission.

10 · Architecture
11 · Design flow
12 · Raw denial rates
13 · DTI and LTV distributions
14 · Case b81d
15 · Review queue
16 · Explain, ask, act
17 · Model sensitivity
18 · SHED and Census context

APPENDIX · ARCHITECTURE

System architecture



DEMOGRAPHICS ARE JOINED ONLY AFTER MATCHING · GRANITE SEES STRUCTURED FACTS ONLY

APPENDIX · DESIGN FLOW

From match to advisory decision

01
Match
Financial twins at the same lender. Finances only, blind to demographics.

→

02
Evidence
Structured case: financial differences, stated reasons, missing information.

→

03
Explain
Reason check against reported data. Granite note from facts only, validated.

→

04
Ask
Reviewer questions answered from the structured evidence only.

→

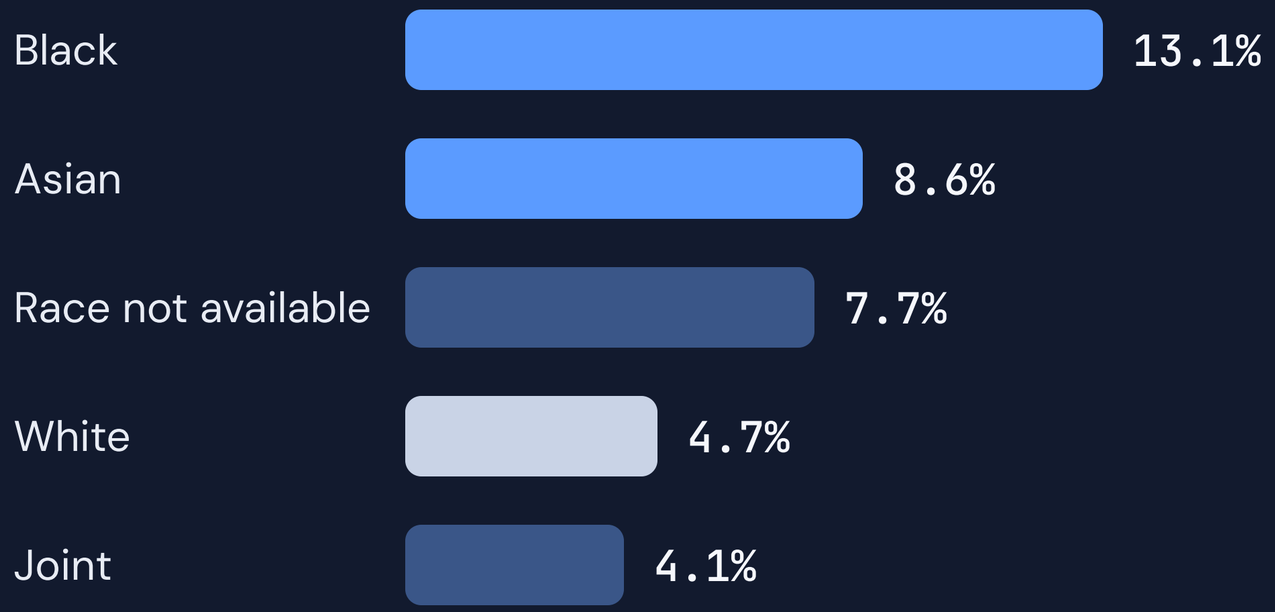
05
Decide
Needs information, no conclusion, or escalate. Logged with the model version.

GUARDRAILS

- Demographics stay in a separate table until after matching
- Granite sees no dollar amounts and no demographics
- Every generated note is validated, with a template fallback
- Advisory only: any other decision is rejected
- Demographic endpoints are token-gated

Before matching, denial rates differ across groups

DENIAL RATE BY APPLICANT GROUP (RAW)

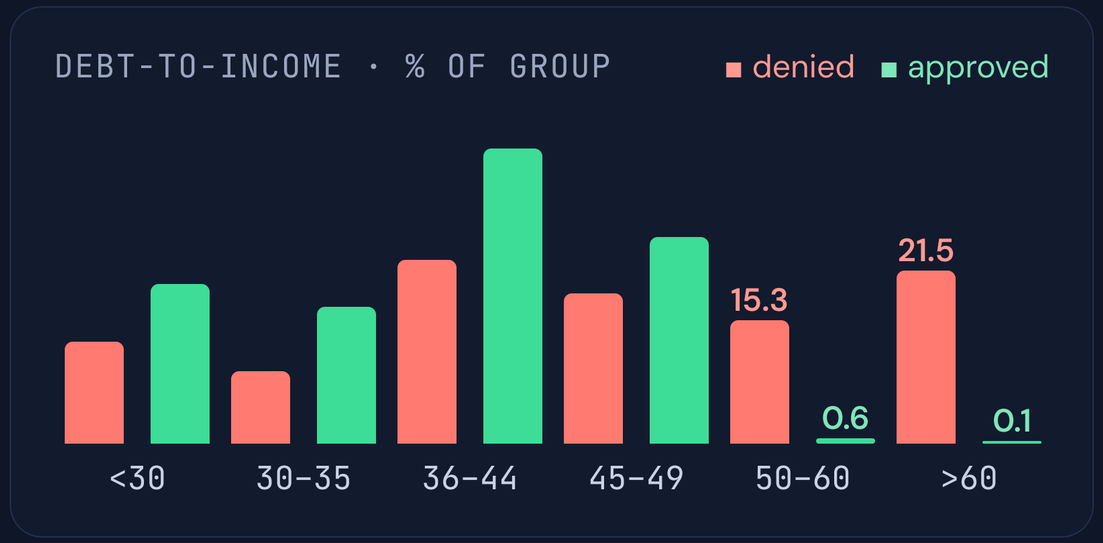


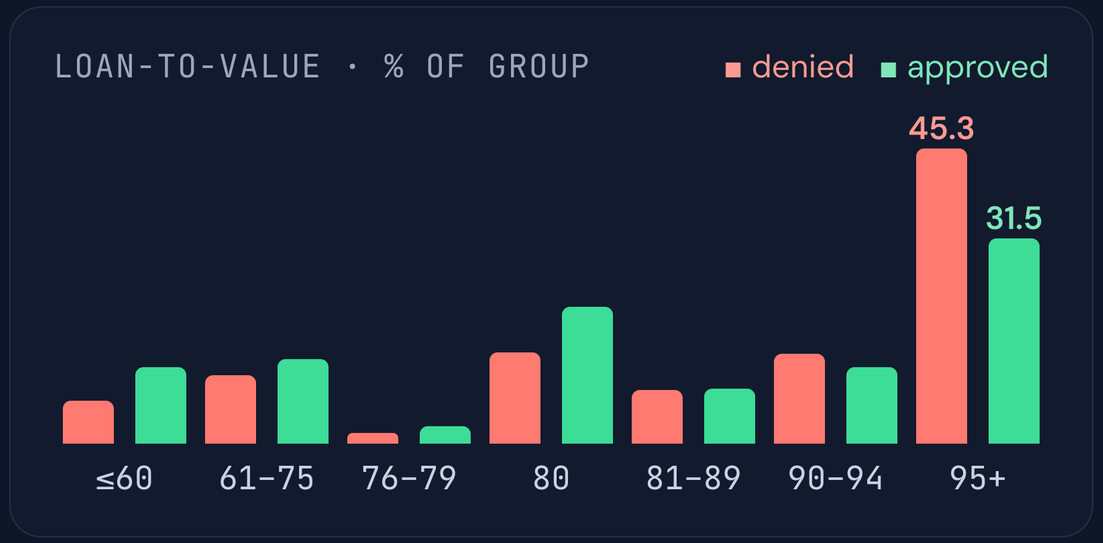
APPLICATIONS (n)	
Black	6,031
Asian	3,884
White	20,339
Joint	1,155
Race not available	5,802

A raw gap is not an adjusted gap

Raw rates mix finances, lenders and places. So SecondLook matches on finances first, then compares.

Finances overlap, except at very high DTI





35.6% of denials have DTI of 50 or more. Only 0.7% of approvals do.

Only 6.5% of denials with DTI above 60 have a comparable approval.

DTI ≥ 50 IS REPORTED AS A BRACKET IN HMDA · 2,600 DENIED, 34,891 APPROVED

One denied case, three approved twins

FIELD	DENIED	TWIN 1	TWIN 2	TWIN 3
Income ($K)	141	141	146	135
Loan amount	$615,000	$615,000	$595,000	$595,000
DTI (%)	46	46	45	43
LTV (%)	95	95	95	95
Property value	$645,000	$645,000	$625,000	$635,000
Distance	—	0.00	0.18	0.23
Match	—	STRONG	STRONG	STRONG
Outcome	DENIED	APPROVED	APPROVED	APPROVED

REASON CHECK

Stated reason: collateral

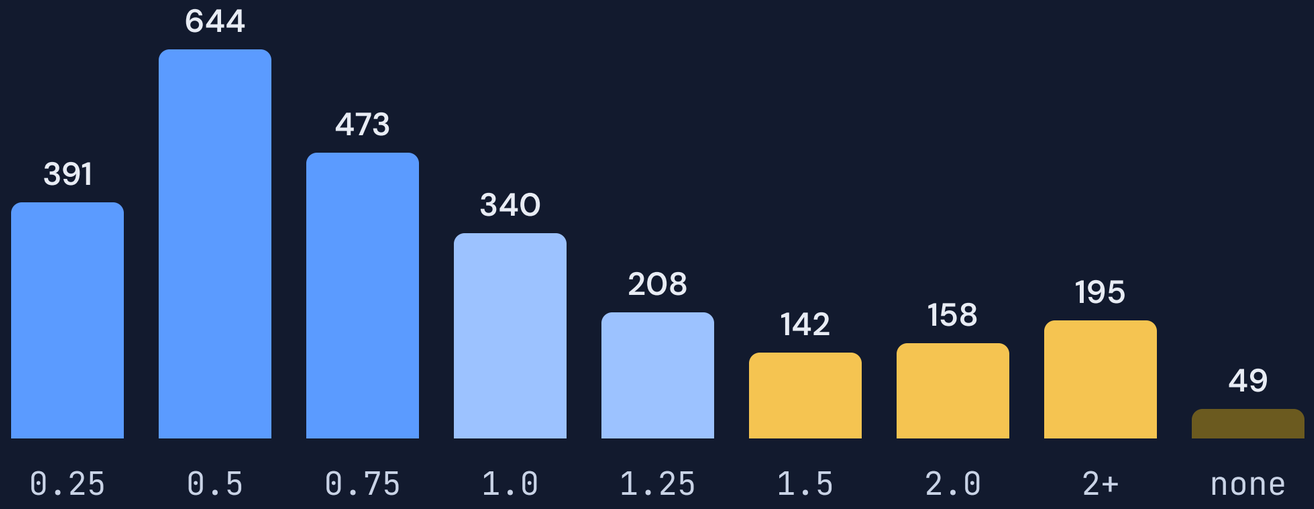
Applicant LTV 95 vs comparators' median 95. Not distinguished by reported data. Appraisal details are not public.

NOT IN PUBLIC DATA

Credit score · assets · employment · documentation · lender policy overlays

So a difference in outcome is a question for review, not a verdict.

Not every denial is forced into the queue



Bins show the upper bound of the average gap, in units of tolerance. Quality also checks the largest single-feature gap.



874 · insufficient comparable evidence

506 of them have DTI above 60%, where almost no approved applications exist. We say so; we never force a flag.

APPENDIX · EXPLAIN · ASK · ACT

AI assists. The reviewer stays in control.

1 · STATED REASON VS DATA

Collateral

Not distinguished by reported data

LTV 95 vs median 95 for approved twins

Every reason gets one of three labels:

✓ Consistent with reported data

≈ Not distinguished by reported data

? Not assessable with public data (credit history, employment)

2 · IBM GRANITE NOTE ✓validated

"This denied application was queued because 3 approved applications from the same lender, out of 1,915 candidates, report closely similar income, loan amount, debt-to-income ratio, loan-to-value ratio, property value and loan term. The stated denial reason was collateral."

Granite sees facts only: no dollar amounts, no demographics. Invented numbers or accusing words are rejected.

3 · ASK, THEN DECIDE

What could explain the denial?

Answers are built from structured results only.

ⓘ Needs additional information

○ Reviewed, no conclusion

▲ **Escalate for further review**

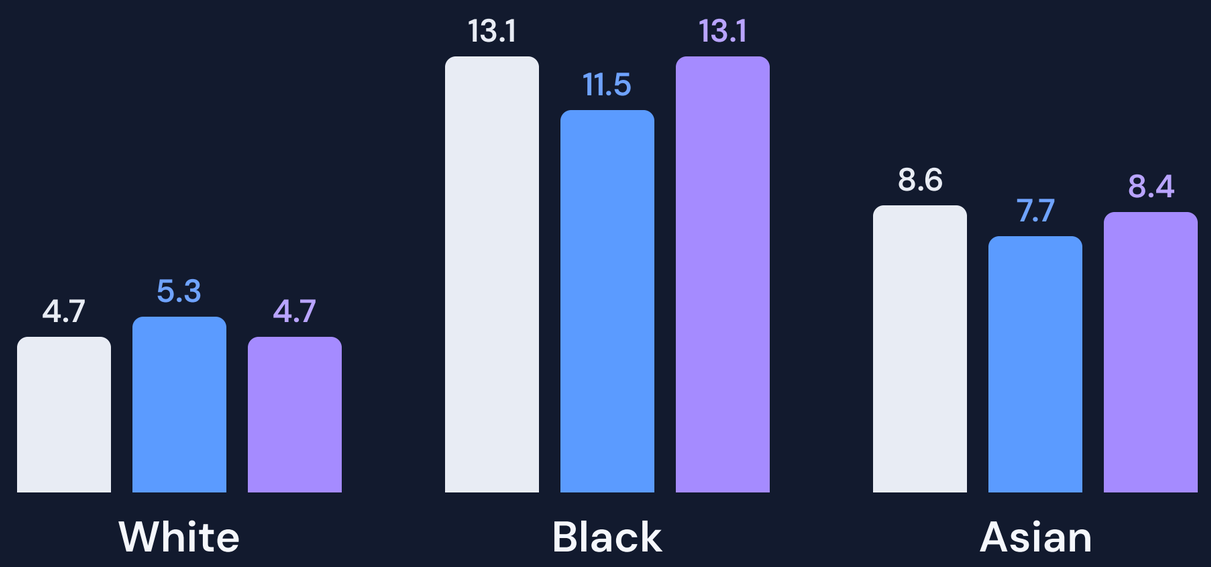
Audit trail: reviewer-1 · v0.1 · strong match

Advisory only. The API rejects any decision except these three (tested).

APPENDIX · MODEL SENSITIVITY · XGBOOST ON LINUXONE

Do the public fields explain the gap? Not fully.

DENIAL RATE (%) · ACTUAL VS MODEL



0.867 → 0.873

AUC without vs with demographics. Overall accuracy barely moves.

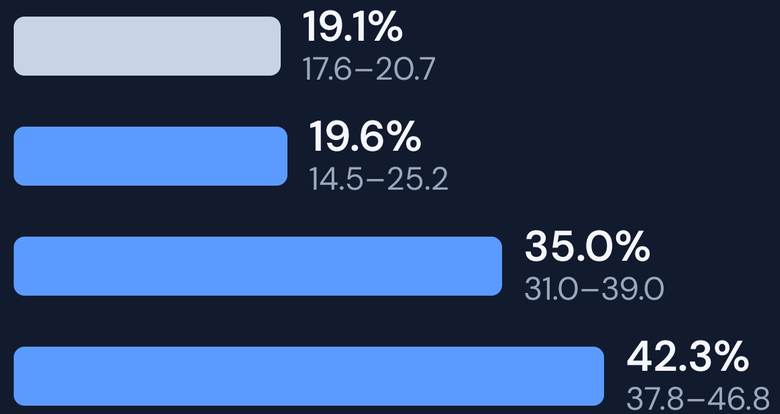
Without demographics the model under-predicts denials for Black applicants (11.5% vs 13.1%). The public finance, lender and county fields do not fully explain group differences.

Predicts historical decisions, not correct ones. Not evidence that a lender used race. The with-demographics model is for auditing only.

The pattern is not only in this dataset

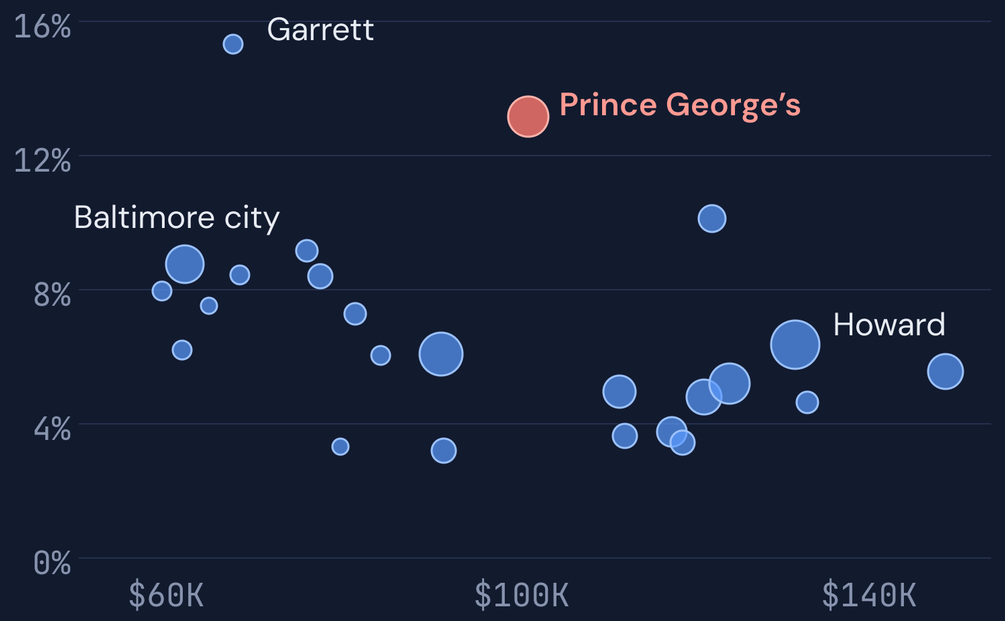
TURNED DOWN FOR CREDIT, PAST 12 MONTHS

US adults who applied for credit · Federal Reserve SHED 2025, survey-weighted



All credit types, not only mortgages. Never joined to HMDA applicants.

MARYLAND COUNTIES · INCOME VS DENIAL RATE



ACS 2024 5-year · dot size = applications · Spearman -0.45 · county level, not individuals